

1、使用环境

1.1、系统：

Ubuntu22.04

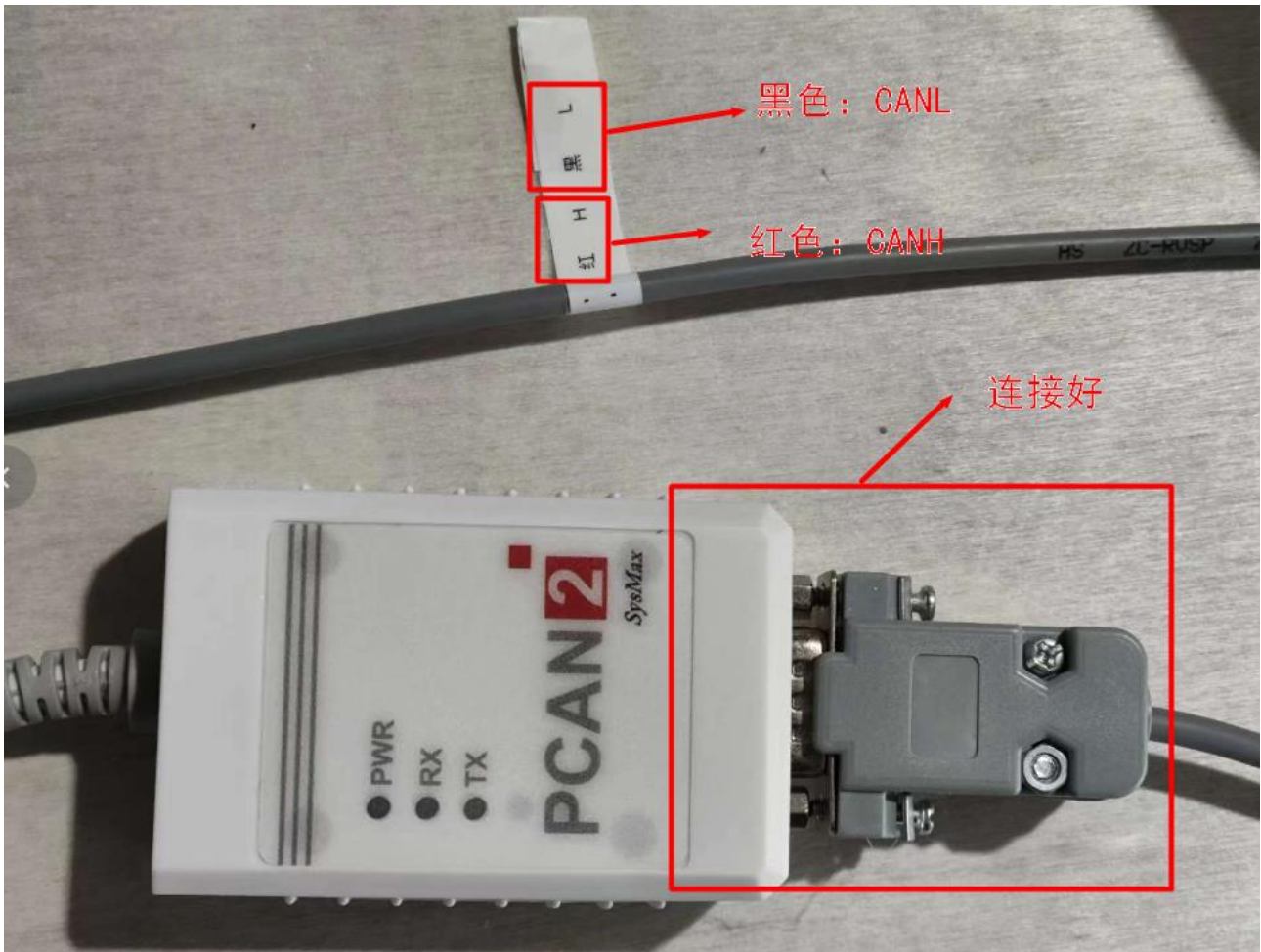
1.2、Ros2 版本：

humble

2、Can 卡连接方式

2.1、Can 卡与底盘连接：

底盘的 CAN-H 连接到 Can 卡接插头的 CANH，底盘的 CAN-L 连接到 Can 卡接插头的 CANL，如下图：



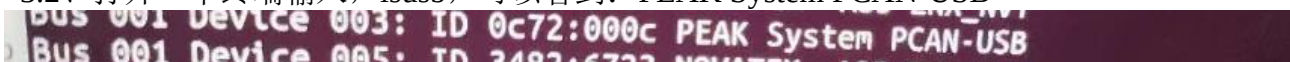
2.2、Can 卡与电脑连接：

usb 线一端直接插到电脑 usb 口

3、Can 卡设置方式：

3.1、Can 卡与电脑连接后的状态，PWR 灯亮

3.2、打开一个终端输入，lsusb，可以看到：PEAK System PCAN-USB



3.3、接着输入，`sudo ip link set can0 up type can bitrate 500000`，这时候，底盘处于上电状态，那么蓝色 RX 灯会闪烁，同时红色 TX 灯也会亮，RX 灯闪烁表示 Can 卡接收到数据，TX 灯闪烁表示发送数据：

4、数据查看

4.1、安装工具：

```
xwqf@ros2:~$ sudo apt install can-utils
```

4.2、查看：

```
xwqf@ros2:~$ candump can0
```

```
can0 18C4D7EF [8] 00 00 97 4A 05 00 50 88
can0 18C4D9EF [8] 00 00 26 5E 05 00 50 2D
can0 18C4D8EF [8] 00 00 29 5E 05 00 50 22
can0 18C4DCEF [8] 14 00 3C 00 00 00 50 78
can0 18C4DDEF [8] D8 FF D8 FF 00 00 50 50
can0 18C4EAEF [8] 00 00 00 00 00 00 B0 B0
can0 18C4D2EF [8] 00 00 00 00 00 00 50 50
can0 18C4D1EF [8] 00 00 00 00 00 00 60 60
can0 18C4D6EF [8] 00 00 94 4A 05 00 60 BB
can0 18C4D7EF [8] 00 00 97 4A 05 00 60 B8
can0 18C4D9EF [8] 00 00 26 5E 05 00 60 1D
can0 18C4D8EF [8] 00 00 29 5E 05 00 60 12
can0 18C4DCEF [8] 14 00 3C 00 00 00 60 48
can0 18C4DDEF [8] D8 FF D8 FF 00 00 60 60
can0 18C4D2EF [8] 00 00 00 00 00 00 60 60
can0 18C4E2EF [8] 50 00 00 20 0D C8 70 C5
can0 18C4D1EF [8] 00 00 00 00 00 00 70 70
can0 18C4D6EF [8] 00 00 94 4A 05 00 70 AB
can0 18C4D7EF [8] 00 00 97 4A 05 00 70 A8
can0 18C4D9EF [8] 00 00 26 5E 05 00 70 0D
can0 18C4D8EF [8] 00 00 29 5E 05 00 70 02
can0 18C4DCEF [8] 14 00 3C 00 00 00 70 58
can0 18C4DDEF [8] D8 FF D8 FF 00 00 70 70
can0 18C4DAEF [8] 00 11 00 00 00 02 E0 F3
can0 18C4EAEF [8] 00 00 00 00 00 00 C0 C0
can0 18C4E1EF [8] B8 0E A6 FF FC 08 70 6B
can0 18C4D2EF [8] 00 00 00 00 00 00 70 70
can0 18C4D1EF [8] 00 00 00 00 00 00 80 80
can0 18C4D6EF [8] 00 00 94 4A 05 00 80 5B
can0 18C4D7EF [8] 00 00 97 4A 05 00 80 58
can0 18C4D9EF [8] 00 00 26 5E 05 00 80 FD
can0 18C4D8EF [8] 00 00 29 5E 05 00 80 F2
can0 18C4DCEF [8] 14 00 3C 00 00 00 80 A8
can0 18C4DDEF [8] D8 FF D8 FF 00 00 80 80
can0 18C4EAEF [8] 00 00 00 00 00 00 D0 D0
```

5、程序运行

5.1、文件存放：

```
xwqf@ros2:~/ros2_ws$ tree -L 3
.
├── src
│   └── mk-01
│       ├── yhs_can_control
│       └── yhs_can_interfaces
└── 4 directories, 0 files
xwqf@ros2:~/ros2_ws$
```

5.2、编译：

```
xwqf@ros2:~/ros2_ws$ colcon build
Starting >>> yhs_can_interfaces
Finished <<< yhs_can_interfaces [10.3s]
Starting >>> yhs_can_control
Finished <<< yhs_can_control [16.4s]

Summary: 2 packages finished [27.0s]
xwqf@ros2:~/ros2_ws$
```

5.3、设置环境变量：

```
xwqf@ros2:~/ros2_ws$ source install/setup.bash
xwqf@ros2:~/ros2_ws$
```

5.4、运行：

```
xwqf@ros2:~/ros2_ws$ ros2 launch yhs_can_control yhs_can_control.launch.py
```

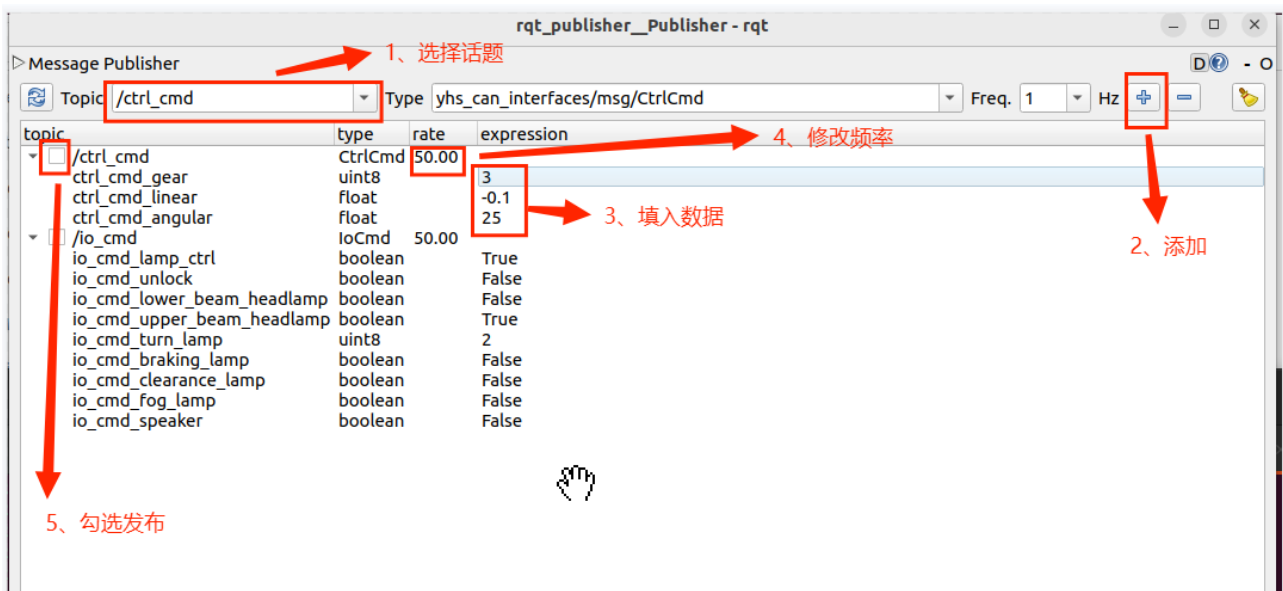
终端打印输出这一行后，再无打印输出，表示数据接收正常：

```
[yhs_can_control_node-1] [INFO] [1679888512.597938093] [yhs_can_control_node]: yhs_can_control_node initialized successfully
```

6、数据发送

6.1、使用 rqt_publisher 工具，遥控器要切换到指令控制模式，建议把底盘架起来测试：

```
xwqf@ros2:~/ros2_ws$ source install/setup.bash
xwqf@ros2:~/ros2_ws$ ros2 run rqt_publisher rqt_publisher
```



7、话题说明

7.1、速度控制，发布频率要在 30hz 以上

消息类型	yhs_can_interfaces/msg/CtrlCmd	
话题名	ctrl_cmd	
变量	类型	说明
ctrl_cmd_gear	uint8	目标档位 00: disable 01: 驻车档 02: 空挡 03: 运动学控制档
ctrl_cmd_linear	float32	目标线速度 单位: m/s
ctrl_cmd_angular	float32	目标角速度 单位: ° /s

7.2、IO 控制，有些是预留的，底盘可能没有此功能

消息类型	yhs_can_interfaces/msg/IoCmd	
话题名	io_cmd	
变量	类型	说明
io_cmd_lamp_ctrl	bool	灯控制权模式 0 = 根据状态自动控制 1 = 自由控制
io_cmd_unlock	bool	安全停车解锁开关 0 = 无效 1 = 解锁使能
io_cmd_lower_beam_headlamp	bool	近光灯开关 0 = 关闭 1 = 打开

io_cmd_upper_beam_headlamp	bool	远光灯开关 0 = 关闭 1 = 打开
io_cmd_turn_lamp	uint8	转向灯开关 0 = 全关 1 = 开启左转向灯 2 = 开启右转向灯
io_cmd_braking_lamp	bool	制动灯开关 0 = 关闭 1 = 打开
io_cmd_clearance_lamp	bool	示廓灯开关 0 = 关闭 1 = 打开
io_cmd_fog_lamp	bool	雾灯开关 0 = 关闭 1 = 打开
io_cmd_speaker	bool	扬声器开关 0 = 关闭 1 = 打开

7.3，底盘数据订阅，有些是预留的，底盘可能没有此项数据反馈

消息类型	yhs_can_interfaces/msg/ChassisInfoFb	
话题名	chassis_info_fb	
变量	类型	说明
ctrl_fb.ctrl_fb_target_gear	uint8	当前档位反馈 00: disable 01: 驻车档 02: 空挡 03: 运动学控制档 04: 自由控制档
ctrl_fb.ctrl_fb_linear	float32	当前车体线速度反馈 单位: m/s
ctrl_fb.ctrl_fb_angular	float32	当前车体角速度反馈 单位: ° /s
io_fb.io_fb_lamp_ctrl	bool	灯控制权状态反馈 0 = 根据状态自动控制 1 = 自由控制
io_fb.io_fb_unlock	bool	安全停车解锁状态反馈 0 = 无效 1 = 解锁成功
io_fb.io_fb_lower_beam_headlamp	bool	近光灯开关状态反馈 0 = 关闭 1 = 打开
io_fb.io_fb_upper_beam_headlamp	bool	远光灯开关状态反馈 0 = 关闭 1 = 打开
io_fb.io_fb_turn_lamp	uint8	转向灯开关状态反馈 0 = 全关 1 = 开启左转向灯 2 = 开启右转向灯
io_fb.io_fb_braking_lamp	bool	制动灯开关状态反馈 0 = 关闭 1 = 打开

io_fb.io_fb_clearance_lamp	bool	示廓灯开关状态反馈 0 = 关闭 1 = 打开
io_fb.io_fb_fog_lamp	bool	雾灯开关状态反馈 0 = 关闭 1 = 打开
io_fb.io_fb_speaker	bool	扬声器开关状态反馈 0 = 关闭 1 = 打开
io_fb.io_fb_fl_impact_sensor	bool	前左防撞条开关状态反馈 0 = 关闭 1 = 打开
io_fb.io_fb_fm_impact_sensor	bool	前中防撞条开关状态反馈 0 = 关闭 1 = 打开
io_fb.io_fb_fr_impact_sensor	bool	前右防撞条开关状态反馈 0 = 关闭 1 = 打开
io_fb.io_fb_rl_impact_sensor	bool	后左防撞条开关状态反馈 0 = 关闭 1 = 打开
io_fb.io_fb_rm_impact_sensor	bool	后中防撞条开关状态反馈 0 = 关闭 1 = 打开
io_fb.io_fb_rr_impact_sensor	bool	后右防撞条开关状态反馈 0 = 关闭 1 = 打开
io_fb.io_fb_fl_drop_sensor	bool	前左跌落传感器状态反馈 0 = 关闭 1 = 打开
io_fb.io_fb_fm_drop_sensor	bool	前中跌落传感器状态反馈 0 = 关闭 1 = 打开
io_fb.io_fb_fr_drop_sensor	bool	前右跌落传感器状态反馈 0 = 关闭 1 = 打开
io_fb.io_fb_rl_drop_sensor	bool	后左跌落传感器状态反馈 0 = 关闭 1 = 打开
io_fb.io_fb_rm_drop_sensor	bool	后中跌落传感器状态反馈 0 = 关闭 1 = 打开
io_fb.io_fb_rr_drop_sensor	bool	后右跌落传感器状态反馈 0 = 关闭 1 = 打开
io_fb.io_fb_estop	bool	急停开关状态反馈 0 = 关闭 1 = 打开
io_fb.io_fb_joyypad_ctrl	bool	遥控器控制状态反馈 0 = 关闭 1 = 打开
io_fb.io_fb_charge_state	bool	充电桩状态反馈 0 = 关闭 1 = 打开
l_wheel_fb.l_wheel_fb_velocity	float32	当前左轮速度反馈

		单位： m/s
l_wheel_fb.l_wheel_fb_pulse	int32	当前左轮脉冲数反馈 轮子单圈 N 个脉冲, N=编码器线束*减速比
r_wheel_fb.r_wheel_fb_velocity	float32	当前右轮速度反馈 单位： m/s
r_wheel_fb.r_wheel_fb_pulse	int32	当前右轮脉冲数反馈 轮子单圈 N 个脉冲, N=编码器线束*减速比
bms_flag_fb.bms_flag_fb_soc	uint8	当前剩余电量百分比
bms_flag_fb.bms_flag_fb_single_ov	bool	单体过压保护 0 = 关闭 1 = 打开
bms_flag_fb.bms_flag_fb_single_uv	bool	单体欠压保护 0 = 关闭 1 = 打开
bms_flag_fb.bms_flag_fb_ov	bool	整组过压保护 0 = 关闭 1 = 打开
bms_flag_fb.bms_flag_fb_uv	bool	整组欠压保护 0 = 关闭 1 = 打开
bms_flag_fb.bms_flag_fb_charge_ot	bool	充电过温保护 0 = 关闭 1 = 打开
bms_flag_fb.bms_flag_fb_charge_ut	bool	充电低温保护 0 = 关闭 1 = 打开
bms_flag_fb.bms_flag_fb_discharge_ot	bool	放电过温保护 0 = 关闭 1 = 打开
bms_flag_fb.bms_flag_fb_discharge_ut	bool	放电低温保护 0 = 关闭 1 = 打开
bms_flag_fb.bms_flag_fb_charge_oc	bool	充电过流保护 0 = 关闭 1 = 打开
bms_flag_fb.bms_flag_fb_discharge_oc	bool	放电过流保护 0 = 关闭 1 = 打开
bms_flag_fb.bms_flag_fb_short	bool	短路保护 0 = 关闭 1 = 打开
bms_flag_fb.bms_flag_fb_ic_error	bool	前端检测 IC 错误 0 = 关闭 1 = 打开
bms_flag_fb.bms_flag_fb_lock_mos	bool	软件锁定 MOS 0 = 关闭 1 = 打开
bms_flag_fb.bms_flag_fb_charge_flag	bool	充电标志位 0 = 关闭 1 = 打开
bms_flag_fb.bms_flag_fb_hight_temperature	float32	当前电池最高温度 ℃
bms_flag_fb.bms_flag_fb_low_temperature	float32	当前电池最低温度

		℃
bms_fb.bms_fb_voltage	float32	当前电池电压 单位：V
bms_fb.bms_fb_current	float32	当前电池电流 单位：A
bms_fb.bms_fb_remaining_capacity	float32	当前电池剩余容量 单位：Ah